```
# Generate sample data (50 observations)
set.seed(123)
sample_data <- rnorm(50, mean = 52, sd = 10)

# Hypothesis Testing
t_test <- t.test(sample_data, mu = 50)
print(t_test)

# Maximum Likelihood Estimation
MLE_mean <- mean(sample_data)
MLE_var <- var(sample_data)

cat("MLE of Mean:", MLE_mean, "\n")
cat("MLE of Variance:", MLE_var, "\n")

# Graphical Visualization
hist(sample_data,
probability = TRUE,
main = "Hypothesis Testing: Bell Curve",
xlab = "Values",
col = "lightblue",
border = "black")

curve(dnorm(x,
mean = MLE_mean,
sd = sqrt(MLE_var)),
col = "red",
lwd = 2,
add = TRUE)

abline(v = 50,
col = "blue",
lwd = 2)
```